

MEGR 2156 Assignment: Motor Mount

Instructions:

This week you obtain working knowledge designing a motor mount. You approach the design in two distinct directions, the first is obtained from stress and the second is from deflection using the beam calculations. Students will turn in the assignment through your portfolio albeit CAD files, pictures of sketches, drawing, calculations, writings etc. Design Assignments tab. **Submit a pdf through canvas.** Make sure your assignment is legible and laid out in a structured manner.

- *If applicable, scan your written work and **embed** them in your assignment page in your portfolio.*

Reading:

- Machinery's Handbook
 - Beams: Beam Calculations: p 256 - 274

Documentation:

The documentation's intent is to capture your work and learning process from the time you read through the assignment all the way until you turn in your work. Your work may include but not limited to your thoughts, your insights, your mistakes, your drawings, your calculations etc. A thorough example of documentation can be found [HERE](#).

- Document each step in the process, no step is too small.
- Document the process which includes many pictures with an overview of images.
- Detail any mistakes throughout the process.
- Actual time it took from start to finish.
- Have a Lessons Learned section

Failure to put a [link](#) for downloading your CAD file will result in a 15% deduction to your final grade.

Failure to input *headers* in your assignment will result in a 15% deduction to your final grade.

Description:

Design a motor mount using the (Brushed 24V DC Gear Motor 3.6Kg.cm/46RPM w/ 99.5:1 Planetary Gearbox) [HERE](#) which attaches to the rigid wall **A**. For **both features**, first design for yield strength and then design for a maximum deflection of .30 mm at the free end. You may select [ABS](#), [PETG](#), or [PLA](#) as a motor mount material. When designing the motor mount take into account a safety factor of 3 and neglect the weight of the motor. For steps 1 and 2 draw a **FBD** of the forces and a concept of your design. Research the design of different motor mounts and place the links in an appendix on your page. Make justifiable approximations in your design to simplify your analysis. (ie. use the beam calculations) Follow Appendix B for the initial approach to set up the design analysis.

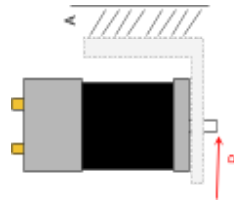


Figure 1: Shows motor, the rigid wall and the force received on the shaft of the motor, where $P = 300 \text{ N}$

In order to get credit for assignment you must use Headers in documentation.

For example

Feature 1

.....

Feature 2

.....

Sketch

.....

CAD Model (Parametric)

.....

Outline:

1. (30%) **Feature 1:** Design a feature of the mount that is attached to the motor. (See appendix B for clarification) Assume the deflection of the feature attached to the wall is zero and the derivative with respect to x is also zero. (aka treat as a cantilever beam) Assume the safety factor accounts for the holes for the motor shaft and the screws in your calculation. Determine the cross sectional area geometry using the beam **bending equations from the lecture**. (Make sure to provide analysis for both stress and deflection)
 - a. (2%) List all the knowns and unknowns.
 - b. (5%) Sketch a FBD of the feature.
 - c. **(20%) Model the equations and symbolically solve.**
 - d. (3%) Numerically solve for the cross-sectional geometry.
2. (30%) **Feature 2:** Design a feature of the mount to be attached to the wall. Assume the rigid wall A can support bolts. Determine the cross sectional geometry using the beam **bending equations discussed (both stress and deflection equations) in lecture**. Choose a material for the mount with enough yield.
 - a. (2%) List all the knowns and unknowns.
 - b. (5%) Sketch a FBD of the feature.
 - c. **(20%) Model the equations and solve them symbolically.**
 - d. (3%) Numerically solve for the cross-sectional geometry.
3. (10%) On paper, sketch the motor mount design as an isometric view. (Determine the dimensions from the previous problems)
4. (30%) Generate a 3D CAD model of the motor mount.
 - a. (6%) Design features on the motor mount to minimize deflection.
 - b. **(20%) Where appropriate, use parametric modeling techniques to design.**
 - c. (4%) Create clearance holes for the shaft and bolts (3.4 mm clearance holes for bolts).

2157 Students Only (20%)

Topic: Drawings

Instructions:

- Using your finished **3D CAD model** of the motor mount, create a **multiview drawing** in CAD.
- Include the **Front View, Right Side View, Top View, and an Isometric View** on the sheet.
- Apply **ASME standard conventions** for:
 - View alignment and placement
 - Line types (hidden lines, centerlines, visible lines)
 - Projection method (third-angle projection)

Dimensions

- **Size dimensions:** Width, height, depth of each part or feature.
- **Location dimensions:** The position of holes, edges, and other features relative to datums or reference geometry.
- **Hole callouts:** Example — *Ø3.4 mm clearance holes for M3 bolts*, using proper annotation format.
- Dimension so that the **part can be manufactured without referencing the CAD model**.
- Avoid redundant or over-dimensioning — follow the “**fully define, but don’t repeat**” rule.

Best Practices

1. **Choose an appropriate scale** so all views are legible without crowding.
2. **Place the isometric view** in the upper-right corner; it’s not dimensioned, but it helps visualize the part.
3. **Add center marks/centerlines** for circular features.

4. **Maintain consistent units** (likely mm, per provided motor dimensions).
5. Include a **title block** with:
 - Your name
 - Date
 - Part name (“Motor Mount”)
 - Scale
 - Material (ABS, PLA, or PETG)
 - Any drawing number or course-specific details your instructor wants.
6. Save/export the drawing as a **PDF** for submission, while also **embedding** it into your portfolio.

Appendix A:

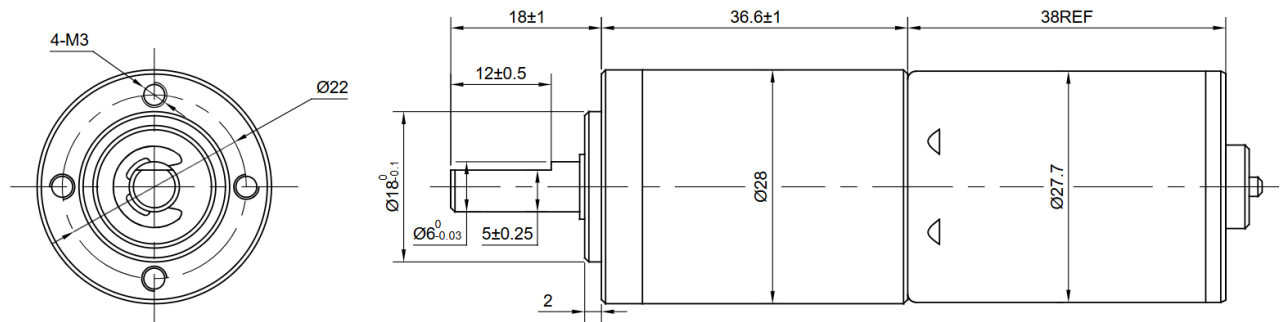
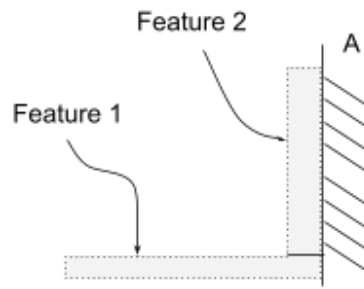
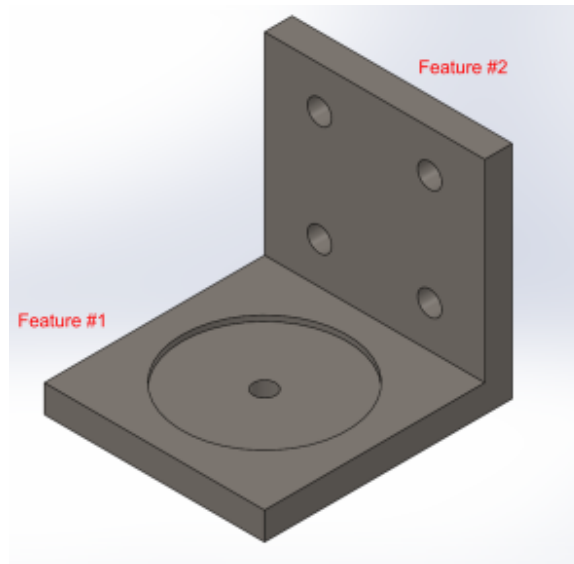


Figure #1 Dimensions of the Gear Motor in mm.

Appendix B:



A 2D representation of a Motor Mount



An isometric view of the motor mount.

